

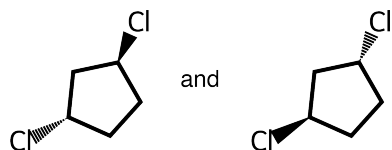
STEREISOMER RELATIONSHIPS — EXAMPLES

ENANTIOMERS

Non-superimposable mirror images; every stereocentre inverted.

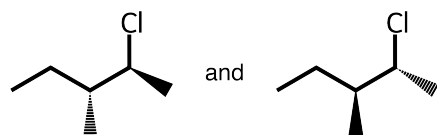
Enantiomers — Wedge-dash formulas

1.



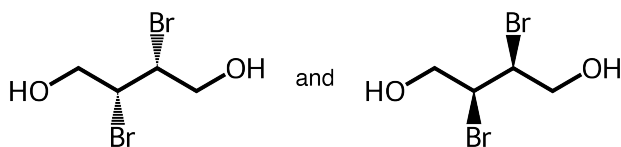
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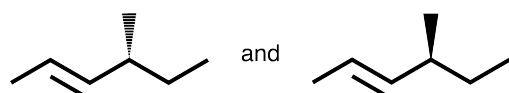
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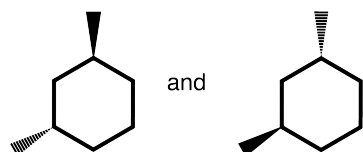
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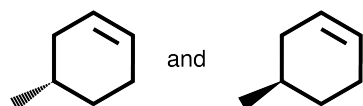
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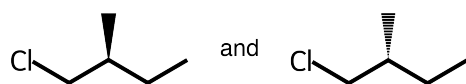
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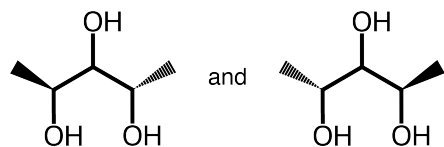
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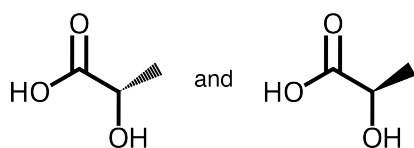
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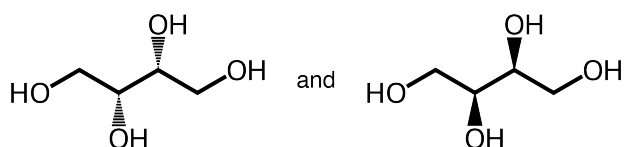
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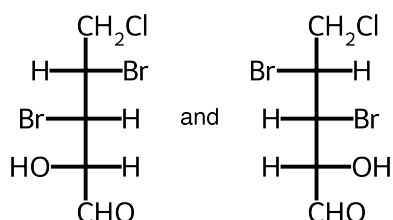
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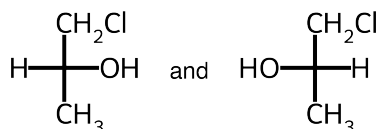
Enantiomers — Fischer projections

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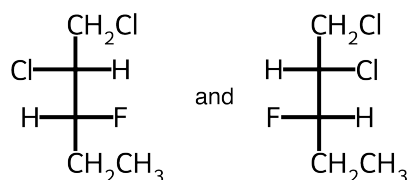
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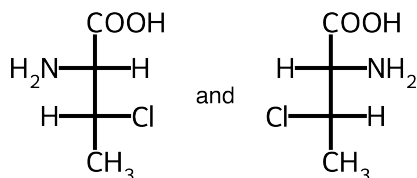
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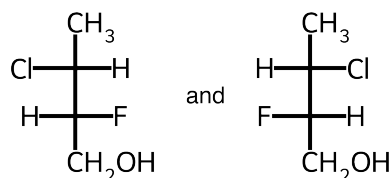
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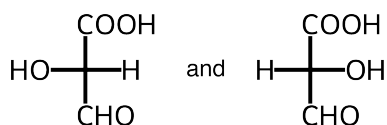
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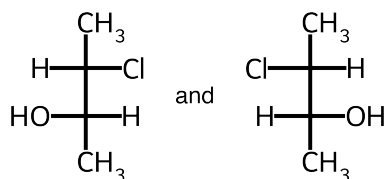
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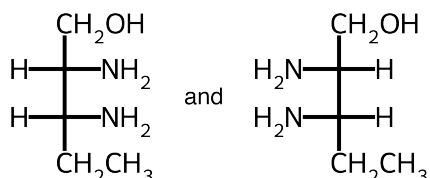
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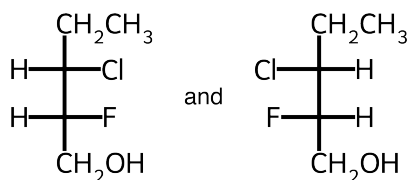
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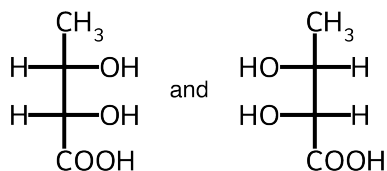
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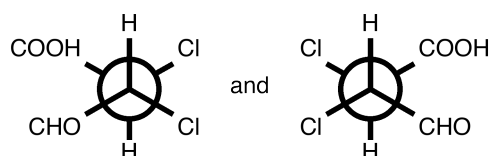
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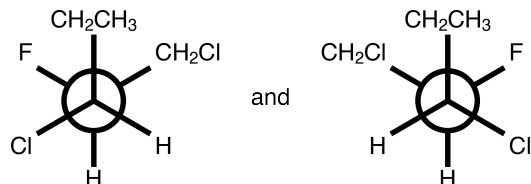
Enantiomers — Newman projections

1.



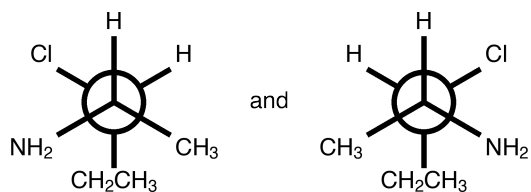
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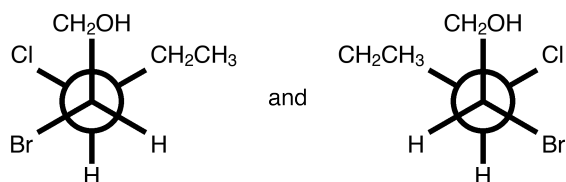
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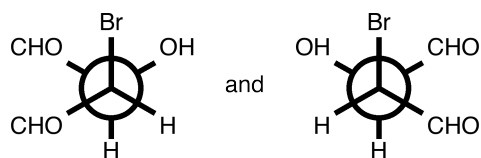
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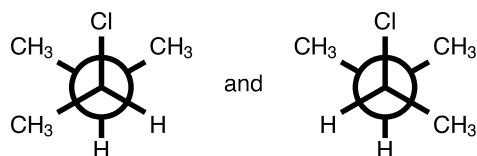
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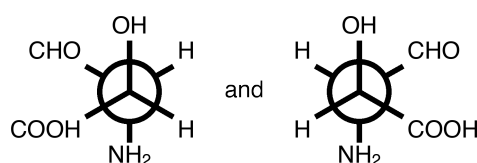
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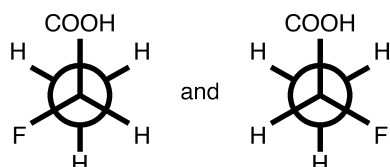
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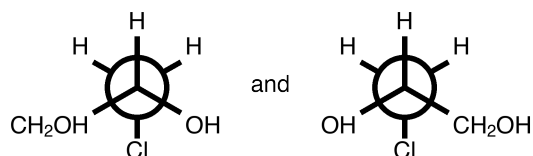
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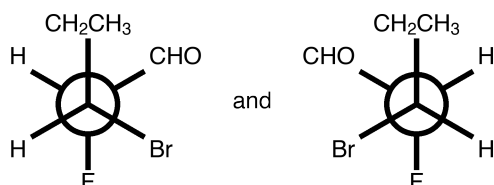
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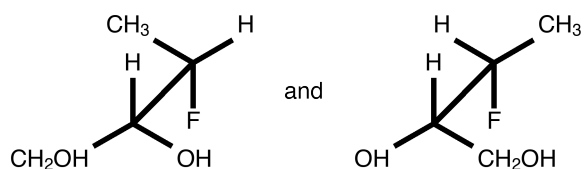
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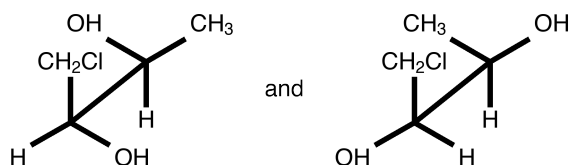
Enantiomers — Sawhorse projections

1.



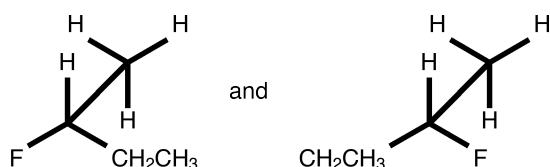
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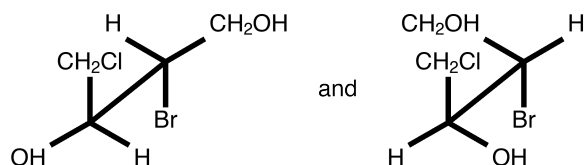
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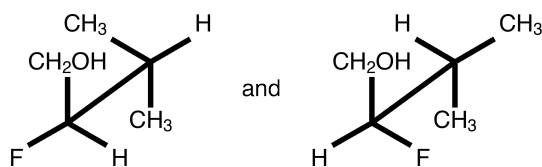
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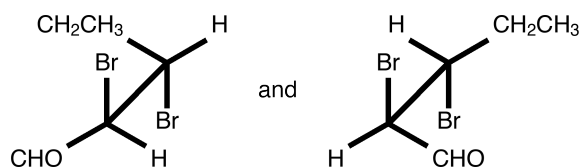
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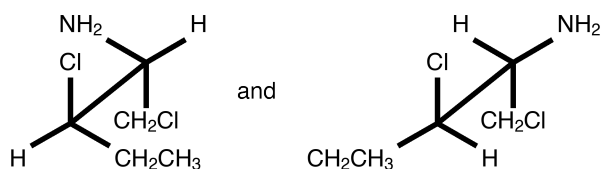
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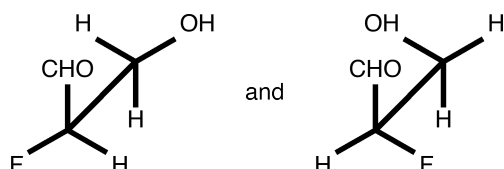
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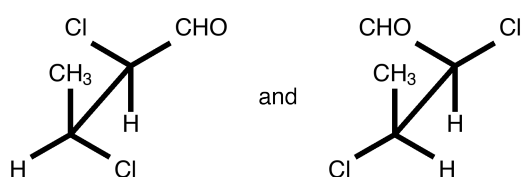
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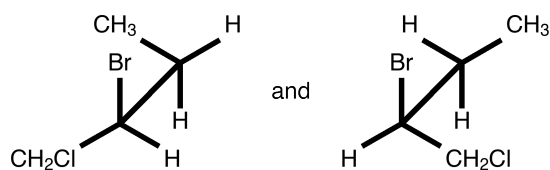
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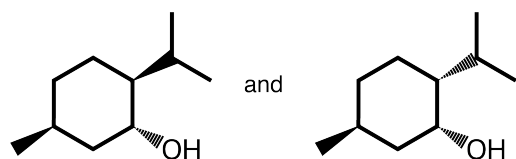
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DIASTEREOMERS

Stereoisomers that are not mirror images; some, but not all, stereocentres inverted.

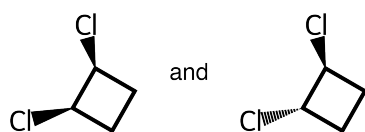
Diastereomers — Wedge-dash formulas

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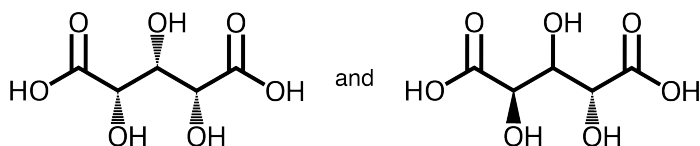
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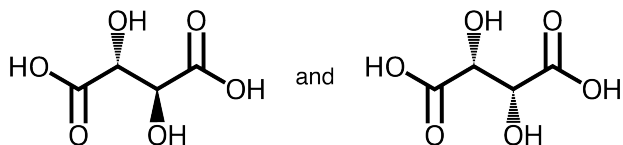
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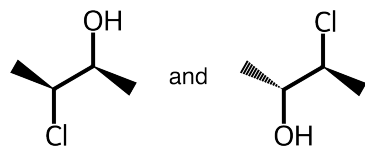
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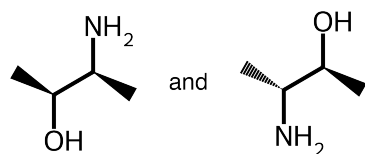
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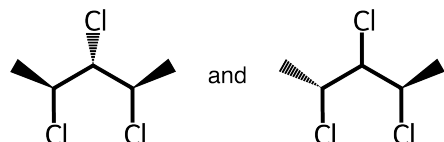
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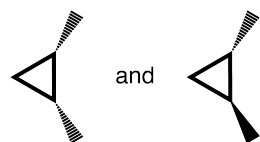
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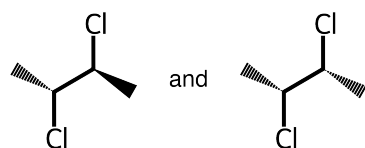
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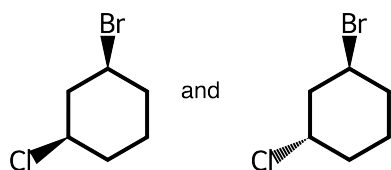
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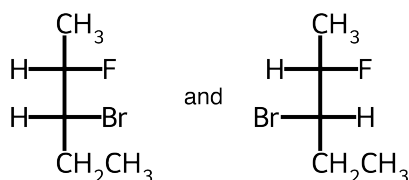
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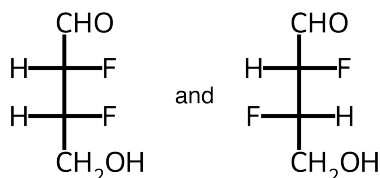
Diastereomers — Fischer projections

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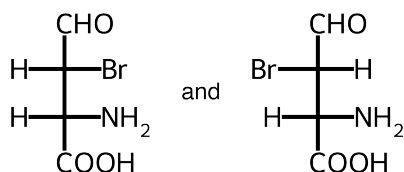
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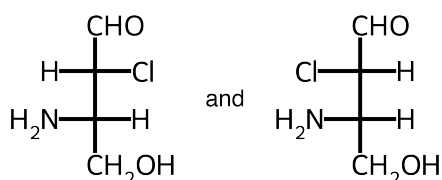
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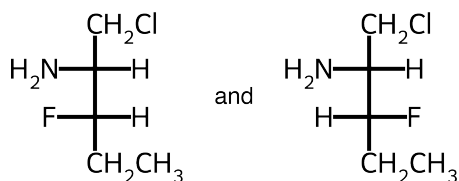
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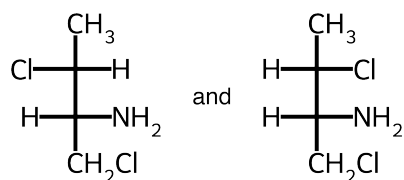
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5.



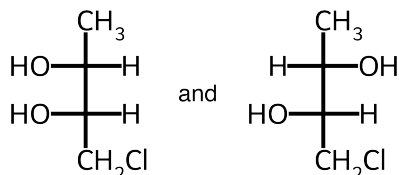
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6.



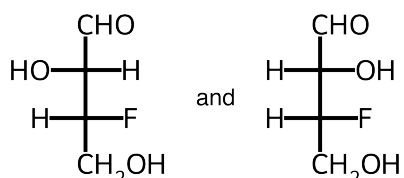
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7.



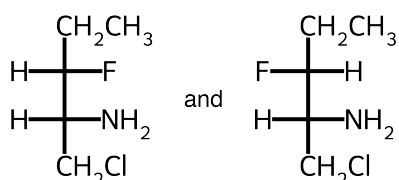
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8.



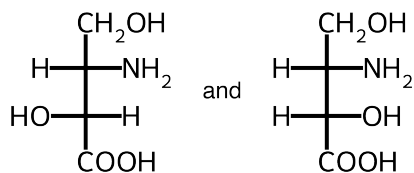
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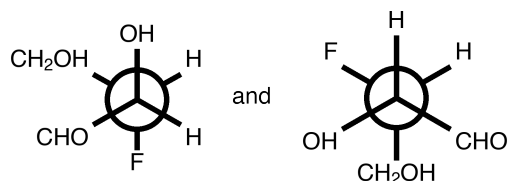
10.



(S,R) and (S,S): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.

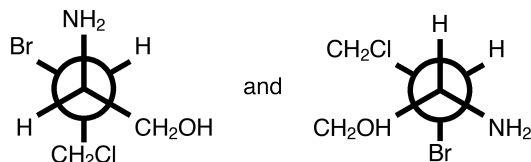
Diastereomers — Newman projections

1.



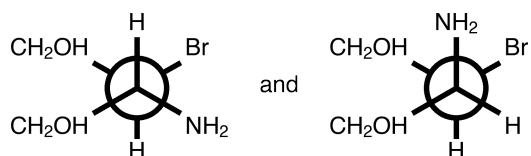
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2.



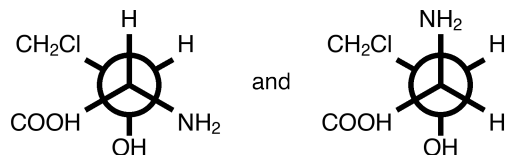
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3.



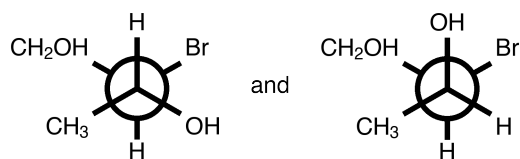
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4.



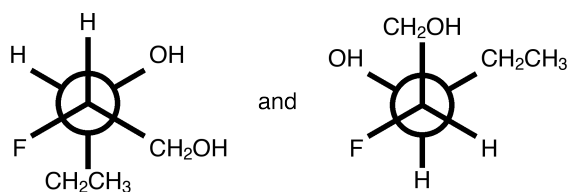
(S,R) and (R,R): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.

5.



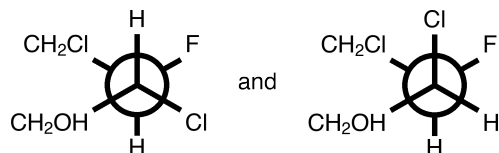
(R,R) and (S,R): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.

6.



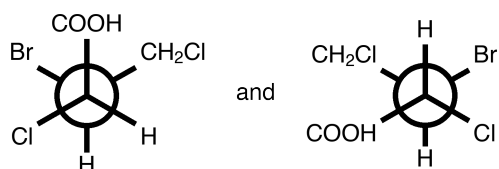
(S,S) and (R,S): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.

7.



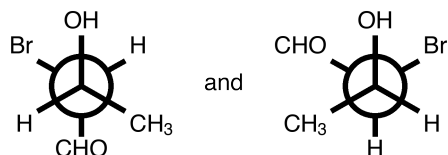
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8.



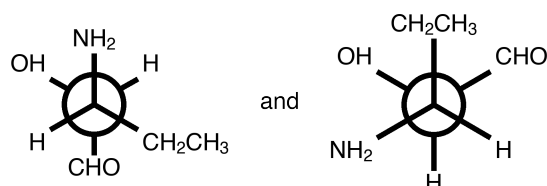
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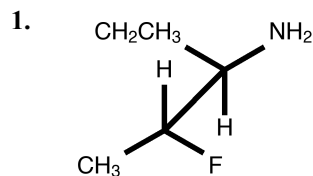
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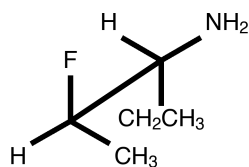


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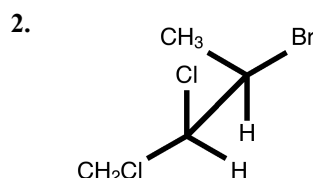
Diastereomers — Sawhorse projections



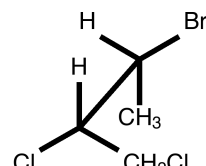
and



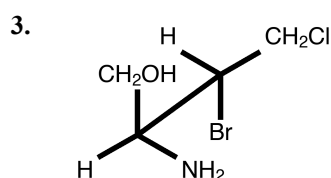
(R,R) and (R,S): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.



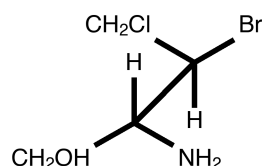
and



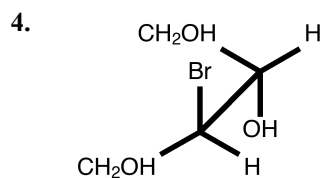
(S,R) and (S,S): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.



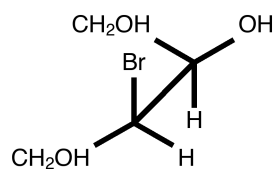
and



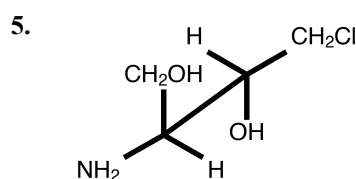
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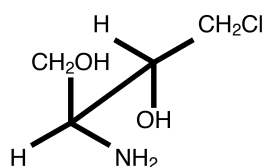
and



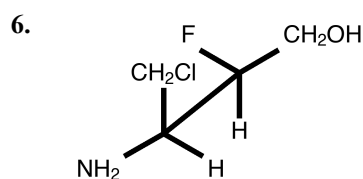
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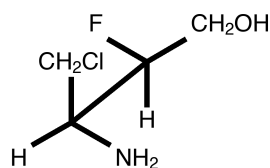
and



(R,S) and (S,S): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.

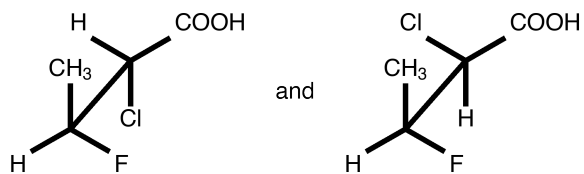


and



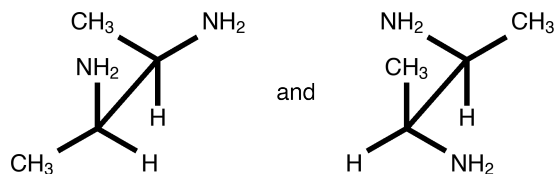
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7.



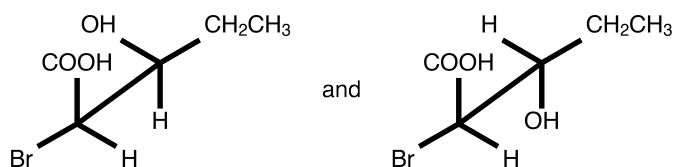
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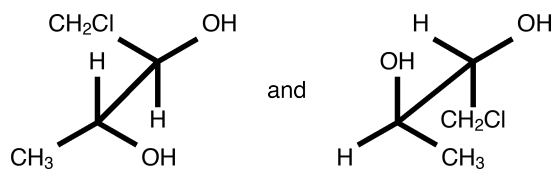
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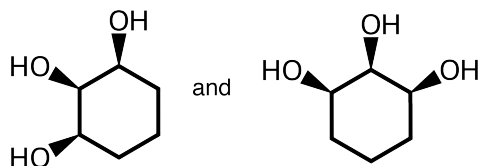
(R,S) and (R,R): the same connectivity but only some of the stereocentres are inverted, so the two are stereoisomers that are not mirror images (diastereomers). They differ in physical properties (bp, mp, solubility) and can be separated by ordinary methods.

HOMOMERS (IDENTICAL COMPOUNDS)

The same molecule drawn in two different ways: all configurations equal.

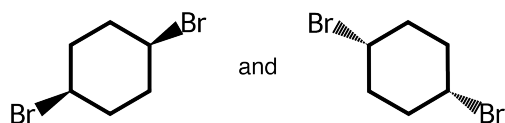
Homomers — Wedge-dash formulas

1.



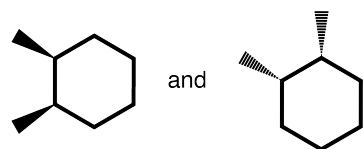
(*s*,*R*,*S*) and (*s*,*R*,*S*): every stereocentre has the same configuration; only the way of drawing differs (the second structure is the same molecule redrawn with the chain turned over / rotated). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

2.



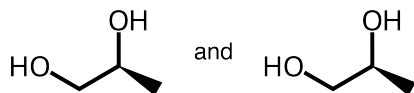
(*s*,*s*) and (*s*,*s*): every stereocentre has the same configuration; only the way of drawing differs (the second structure is the same molecule redrawn with the chain turned over / rotated). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

3.



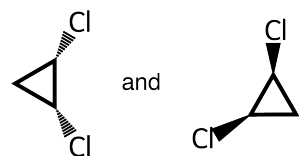
(*R*,*S*) and (*R*,*S*): every stereocentre has the same configuration; only the way of drawing differs (the second structure is the same molecule redrawn with the chain turned over / rotated). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

4.



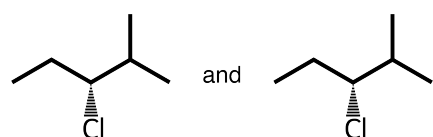
(*S*) and (*S*): every stereocentre has the same configuration; only the way of drawing differs (the second structure is the same molecule redrawn with the chain turned over / rotated). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

5.



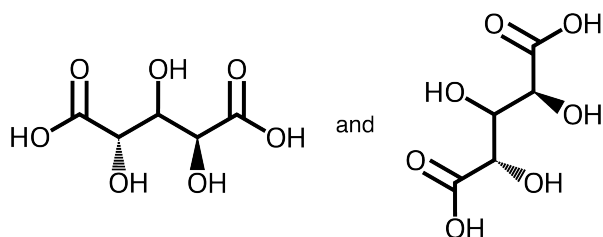
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6.



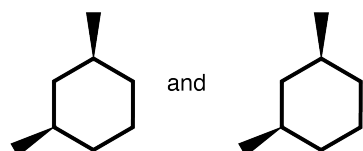
(*R*) and (*R*): every stereocentre has the same configuration; only the way of drawing differs (the second structure is the same molecule redrawn with the chain turned over / rotated). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

7.



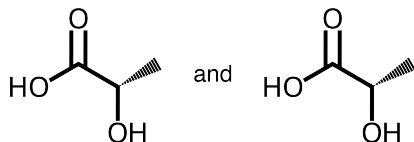
(S,S) and (S,S): every stereocentre has the same configuration; only the way of drawing differs (the second structure is the same molecule redrawn with the chain turned over / rotated). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

8.



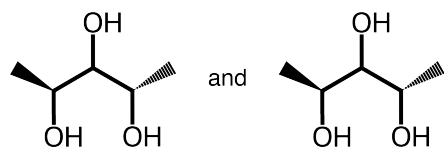
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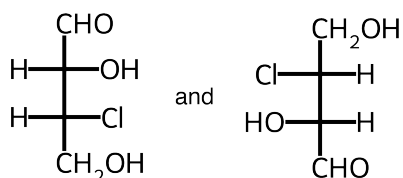
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(S,S) and (S,S): every stereocentre has the same configuration; only the way of drawing differs (the second structure is the same molecule redrawn with the chain turned over / rotated). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

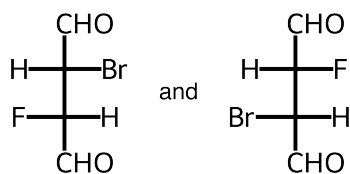
Homomers — Fischer projections

1.



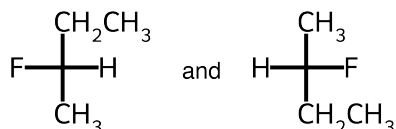
(S,R) and (R,S): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

2.



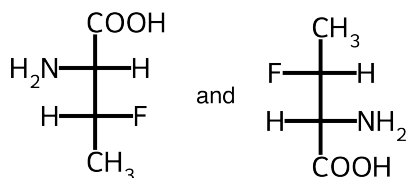
(S,S) and (S,S): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

3.



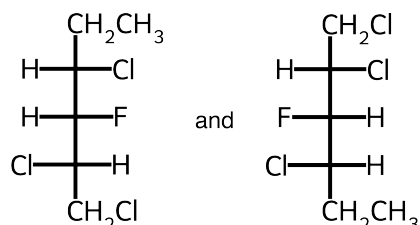
(S) and (S): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

4.



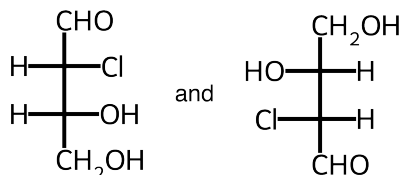
(R,R) and (R,R): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

5.



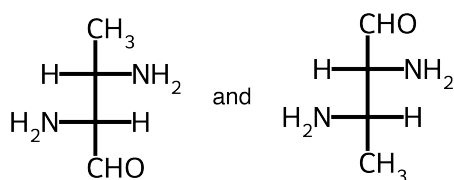
(S,S,R) and (R,S,S): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

6.



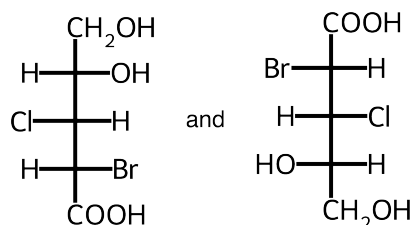
(R,R) and (R,R): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

7.



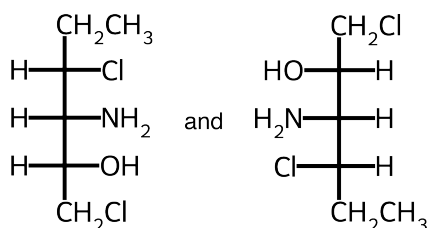
(S,R) and (R,S): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

8.



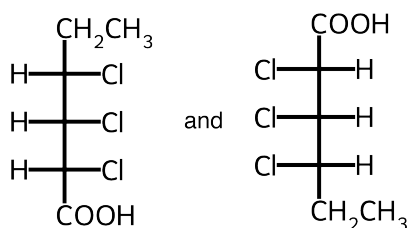
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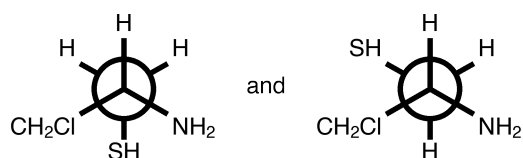
10.



(S,S,R) and (R,S,S): every stereocentre has the same configuration; only the way of drawing differs (the second Fischer projection is the first turned through 180° in the plane of the paper). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

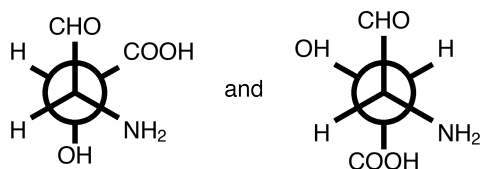
Homomers — Newman projections

1.



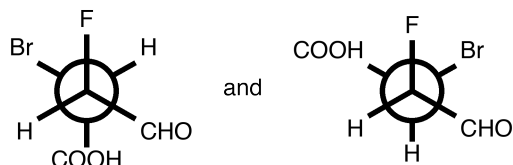
(S) and (S): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

2.



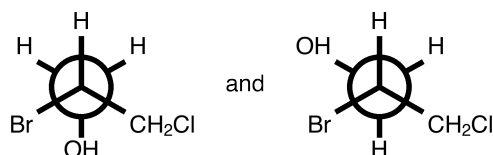
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3.



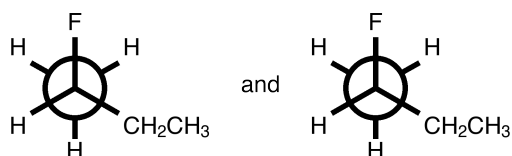
(R,R) and (R,R): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

4.



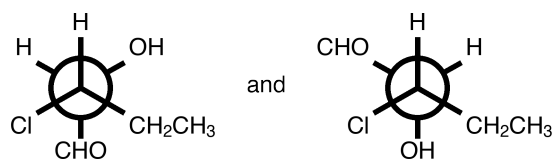
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5.



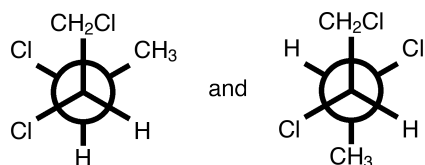
(S) and (S): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

6.



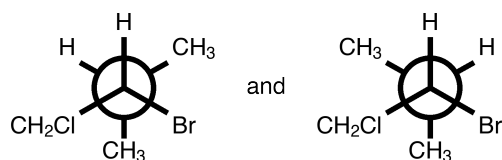
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7.



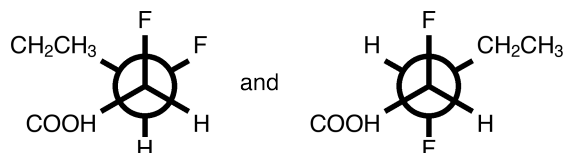
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8.



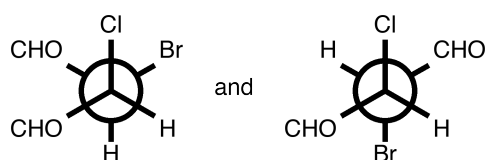
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(S,R) and (S,R): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

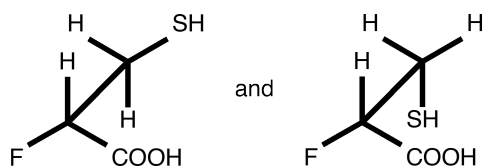
10.



(S,R) and (S,R): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

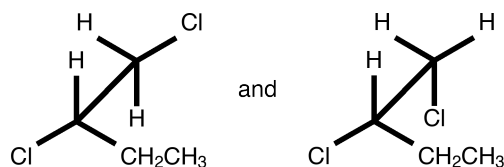
Homomers — Sawhorse projections

1.



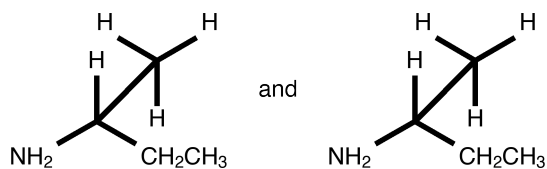
(S) and (S): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

2.



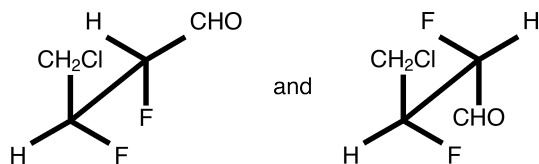
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3.



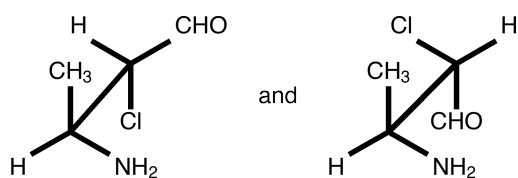
(R) and (R): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

4.



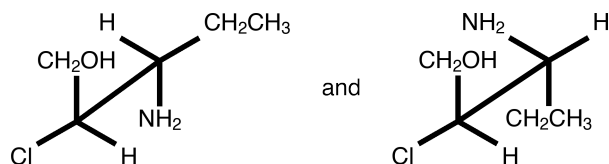
(R,R) and (R,R): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

5.



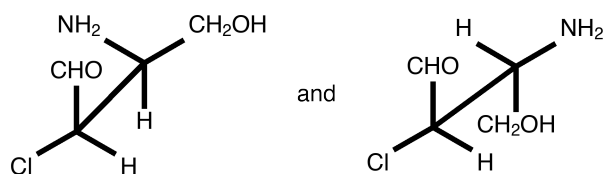
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6.



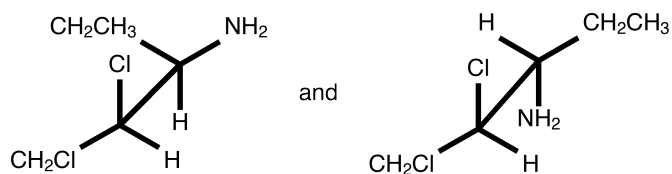
(S,R) and (S,R): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

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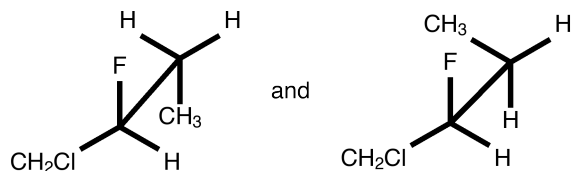
(S,S) and (S,S): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

8.



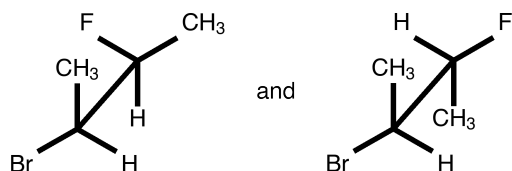
(R,R) and (R,R): every stereocentre has the same configuration; only the way of drawing differs (the second drawing is the first with the rear carbon turned by 120° about the C–C bond (a different conformer)). Rotating or turning one drawing gives the other, so they are the same compound (homomers).

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10.



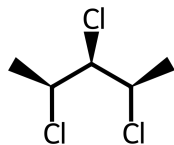
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MESO COMPOUNDS

Contain stereocentres but have an internal plane or centre of symmetry: achiral, optically inactive.

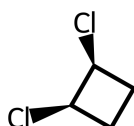
Meso Compounds — Wedge-dash formulas

1.



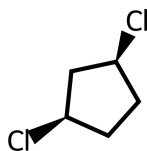
(S,r,R): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

2.



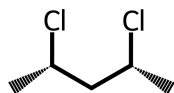
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

3.



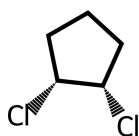
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

4.



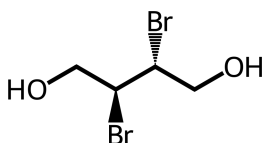
(S,R): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

5.



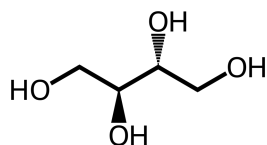
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

6.



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7.



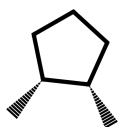
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

8.



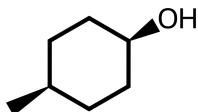
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

9.



(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

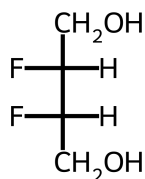
10.



(s,s): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (internal mirror plane / centre). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

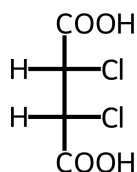
Meso Compounds — Fischer projections

1.



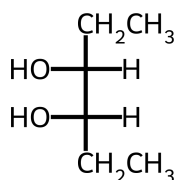
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (a plane of symmetry passes through the middle of the chain in this eclipsed Fischer form). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

2.



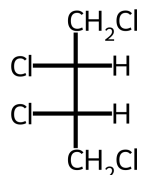
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3.



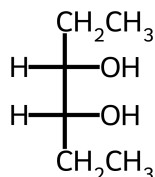
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4.



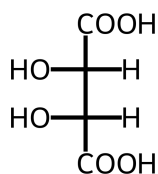
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5.



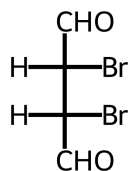
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6.



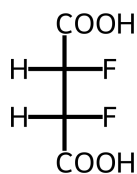
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7.



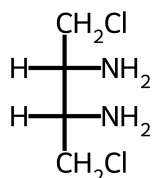
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8.



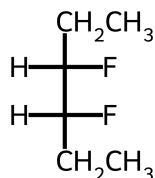
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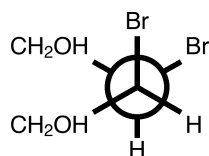
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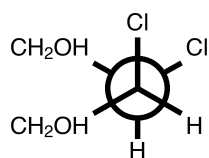
Meso Compounds — Newman projections

1.



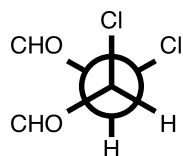
(S,R): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (the front and back carbons carry mirror-related groups; in the anti conformer there is a centre of symmetry, in the eclipsed one a plane). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

2.



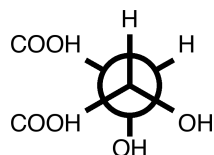
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3.



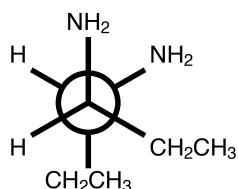
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4.



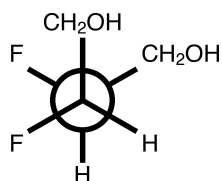
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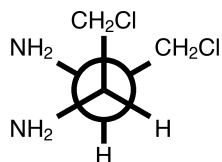
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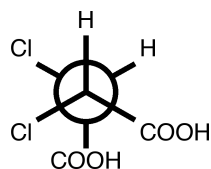
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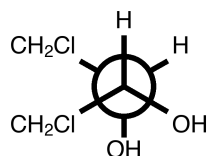
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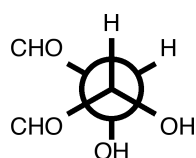
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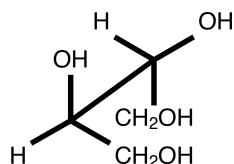
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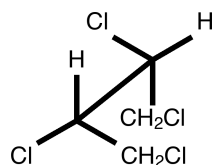
Meso Compounds — Sawhorse projections

1.



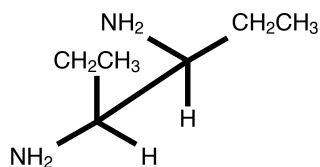
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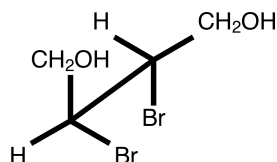
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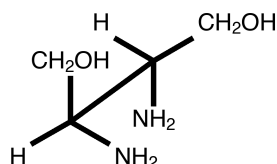
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (the front and back carbons carry mirror-related groups; in the anti conformer there is a centre of symmetry, in the eclipsed one a plane). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

4.



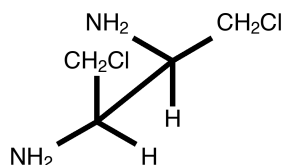
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5.



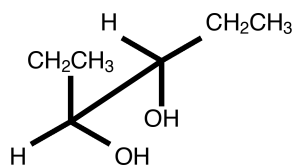
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6.



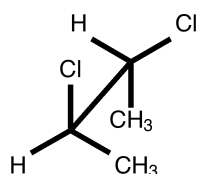
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7.



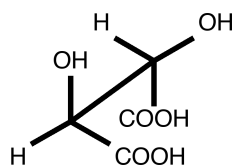
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8.



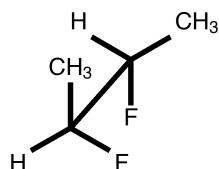
(R,S): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (the front and back carbons carry mirror-related groups; in the anti conformer there is a centre of symmetry, in the eclipsed one a plane). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

9.



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10.

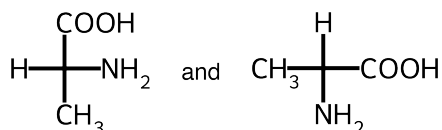


(S,R): the molecule has stereocentres of opposite configuration related by an internal mirror plane or centre of symmetry (the front and back carbons carry mirror-related groups; in the anti conformer there is a centre of symmetry, in the eclipsed one a plane). It is superimposable on its mirror image, therefore achiral and optically inactive although it has stereocentres (meso).

TRICKY EXAMPLES THAT CONFUSE STUDENTS

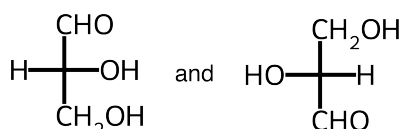
Pairs that look like one relationship but are another.

1. Fischer projection turned through 90°



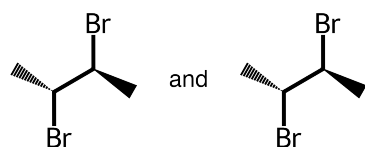
The second projection looks like a harmless rotation of the first, but a Fischer projection may be turned only by 180° in the plane. A 90° turn changes which bonds point towards the viewer, so the configuration is inverted: the pair is ENANTIOMERS.

2. Two swaps of groups in a Fischer projection



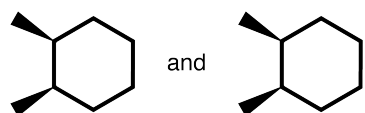
One interchange of two groups inverts a stereocentre, two interchanges restore it. Here the top and bottom groups are exchanged and H and OH are exchanged (an even number of swaps): the result is the original configuration, so the two drawings are HOMOMERS (the second is simply the first turned 180°).

3. meso-2,3-Dibromobutane and its 'mirror image'



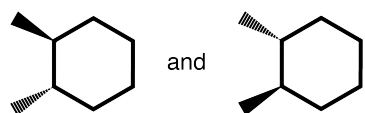
A student draws the mirror image expecting an enantiomer. For a meso compound the mirror image is superimposable on the original (rotate it 180°), so the two drawings are HOMOMERS: meso forms have no enantiomer.

4. cis-1,2-Dimethylcyclohexane and its mirror image



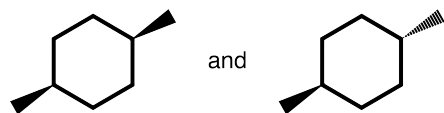
Two stereocentres, both drawn as the 'mirror pair', yet cis-1,2-dimethylcyclohexane has an internal plane of symmetry (it is meso). The mirror image is the same compound: HOMOMERS. Only the TRANS form exists as a pair of enantiomers.

5. trans-1,2-Dimethylcyclohexane and its mirror image



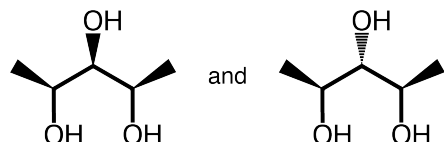
Here no plane of symmetry exists, so the mirror image cannot be superimposed: ENANTIOMERS (trans-1,2-dimethylcyclohexane is chiral), although the cis isomer of the same formula is meso.

6. cis- and trans-1,4-Dimethylcyclohexane



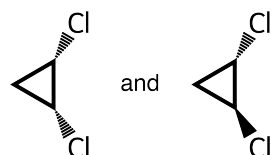
There are no stereocentres here (C1 and C4 carry two identical ring paths), yet cis and trans are different compounds: they are stereoisomers that are not mirror images, i.e. DIASTEREOMERS. Both are achiral, so neither has an enantiomer.

7. Two different meso forms of pentane-2,3,4-triol



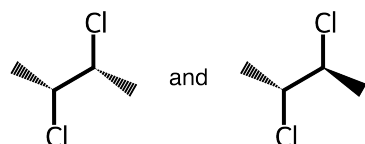
Both compounds are meso (internal mirror plane) and so look like 'the same compound' to many students. They differ only in the configuration of the central pseudoasymmetric carbon ((S,r,R) versus (S,s,R)), so they are DIASTEREOMERS: two meso compounds can be diastereomers of each other.

8. cis- and trans-1,2-Dichlorocyclopropane



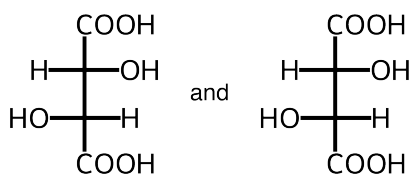
The cis isomer is meso and the trans isomer is chiral (exists as a pair of enantiomers). The relation between cis and either trans enantiomer is DIASTEREOMERS, even though one of them is achiral and the other chiral.

9. (R,R)-2,3-Dichlorobutane and meso-2,3-dichlorobutane



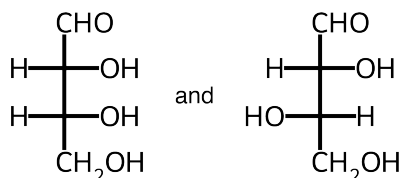
Both have two similar-looking halves and similar labels, but only the meso form has a plane of symmetry. One stereocentre is inverted, not both, so they are DIASTEREOMERS (different melting and boiling points).

10. (R,R)-Tartaric acid: a C₂ axis, not a plane



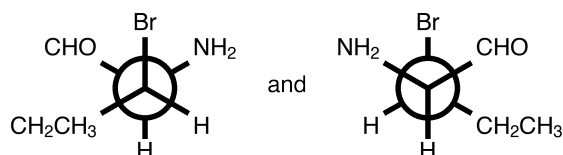
The molecule looks symmetrical, so students call it meso, but the two OH groups lie on OPPOSITE sides. Turning the projection by 180° reproduces it exactly (a C₂ axis), which makes the two drawings HOMOMERS of a CHIRAL compound; meso-tartaric acid has both OH on the same side (a mirror plane).

11. Erythrose and threose



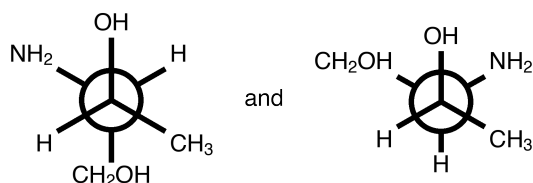
Only one of the two stereocentres is inverted, so the pair is **DIASTEREOMERS**, not enantiomers (the enantiomer of the left structure would need BOTH centres inverted).

12. Newman projection viewed from the other end



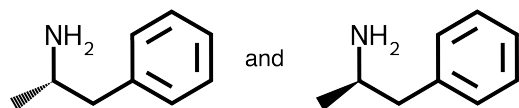
The second drawing looks quite different because the molecule is viewed along the same C–C bond from the opposite end (front and back carbons exchanged, left and right reversed). The molecule is unchanged: **HOMOMERS**.

13. Two staggered conformers of the same molecule



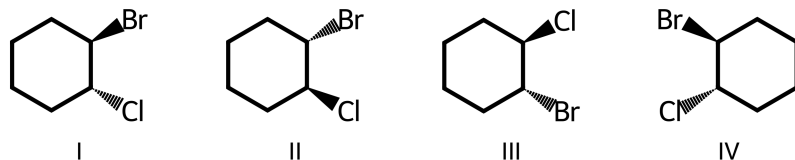
The two drawings look different (different groups are anti or gauche), but they differ only by rotating the rear carbon through 120° about the C–C single bond: they are conformers, i.e. the same compound (**HOMOMERS**), not separate isomers.

14. Practice 1 – A drawing and its reflection in a mirror line



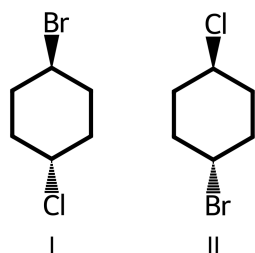
Reflect a drawn structure in a vertical or a horizontal mirror line lying in the plane of the paper: a wedge stays a wedge, but the molecule becomes its mirror image. A reflected drawing is therefore an **ENANTIOMER** of the original (not the same compound); only turning the paper (rotation in the plane, keeping wedges as wedges) gives the same molecule.

15. Practice 2 – 1-Bromo-2-chlorocyclohexane in four drawings



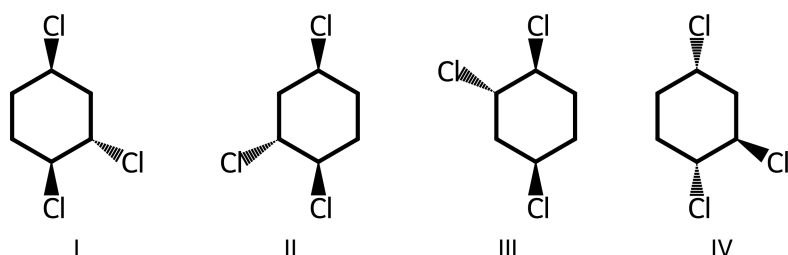
I–II: **ENANTIOMERS**; I–III: **IDENTICAL**; I–IV: **ENANTIOMERS**; II–III: **ENANTIOMERS**; II–IV: **IDENTICAL**; III–IV: **ENANTIOMERS**. Drawing III is I turned over (180° about a horizontal in-plane axis, which swaps wedge and dash and top and bottom), so I and III are the same compound; IV is the left–right mirror image of I, hence identical to II.

16. Practice 3 – trans-1-Bromo-4-chlorocyclohexane



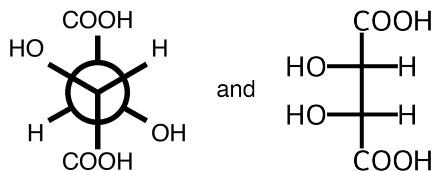
I-II: IDENTICAL. The trans-1,4 compound has a plane of symmetry, so its mirror image (which is what exchanging the two drawings produces) is the same molecule.

17. Practice 4 – 1,2,4-Trichlorocyclohexane in four drawings



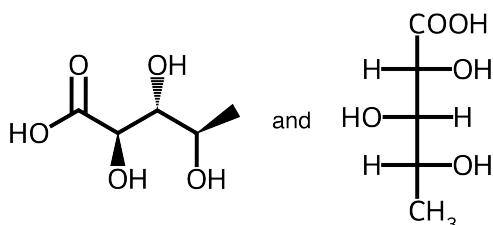
I-II: ENANTIOMERS; I-III: IDENTICAL; I-IV: ENANTIOMERS; II-III: ENANTIOMERS; II-IV: IDENTICAL; III-IV: ENANTIOMERS. Compare each pair by checking whether every wedge/hash is inverted (mirror image) or the second drawing is just the first turned over or rotated.

18. Practice 5 – Newman projection against Fischer projection



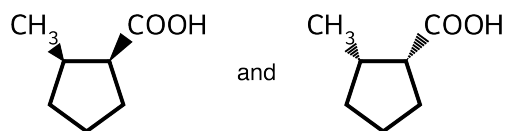
Both are meso-tartaric acid (IDENTICAL): in the Fischer projection both OH groups are on the same side (a plane of symmetry); in the staggered Newman projection the two OH groups are anti, which gives a centre of symmetry. The molecule is the same.

19. Practice 6 – Zig-zag wedge formula against Fischer projection



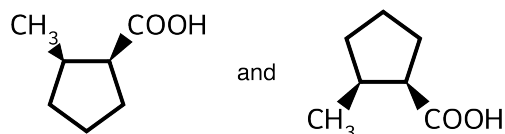
Convert both to the same representation and compare the three centres: DIASTEREOMERS (one or two, but not all three, centres differ). Always convert zig-zag to Fischer (or the reverse) before deciding.

20. Practice 7 – cis-2-Methylcyclopentane-1-carboxylic acid: all groups inverted



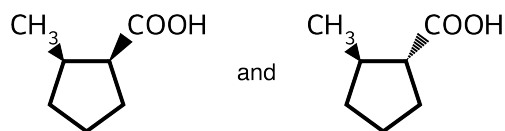
a-b: ENANTIOMERS. Inverting every wedge to a hash gives the mirror image.

21. Practice 8 – the same structure drawn upside down



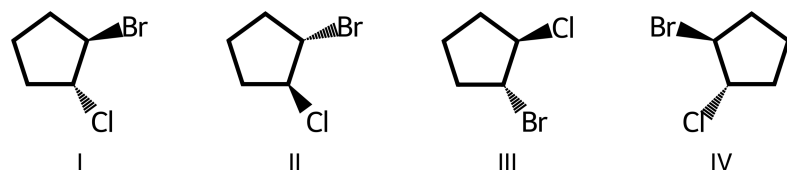
a-b: ENANTIOMERS. Turning the ring upside down by reflecting the drawing (wedges stay wedges) is a mirror image; turning it by rotating the paper would have been the same compound.

22. Practice 9 – cis against trans



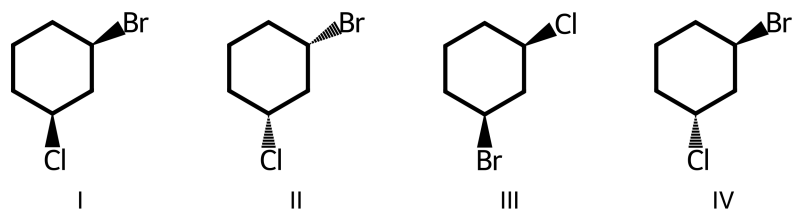
a-b: DIASTEREOMERS. Only one centre is inverted: cis and trans are diastereomers.

23. Practice 10 – 1-Bromo-2-chlorocyclopentane in four drawings (same style as Practice 2)



I-II: ENANTIOMERS; I-III: IDENTICAL; I-IV: ENANTIOMERS; II-III: ENANTIOMERS; II-IV: IDENTICAL; III-IV: ENANTIOMERS

24. Practice 11 – 1-Bromo-3-chlorocyclohexane: cis and trans drawings



I-II: ENANTIOMERS; I-III: ENANTIOMERS; I-IV: DIASTEREOMERS; II-III: IDENTICAL; II-IV: DIASTEREOMERS; III-IV: DIASTEREOMERS. The cis-1,3 compound is chiral here because Br and Cl differ: its mirror image has all wedges inverted.